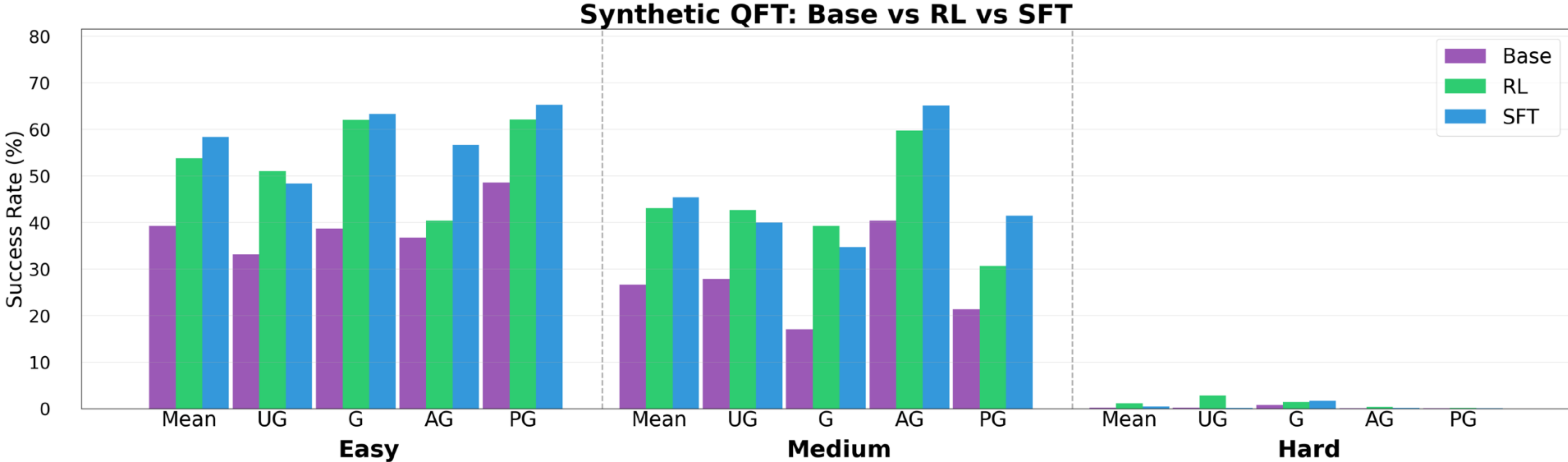


Domain vs Operational Difficulty



No consistent correlation between pass rate & domain difficulty

Fine-Tuning in Academia

Compute Challenges of RL

- Our simple RL training consumed ~**160** gpu hours (all studies ~**240 GPU days**)
- Extending performance **purely with outcome RL** not feasible in academia

Promising Directions for Fine-Tuning in Academia

- Domain specific SFT avoids computational costs
- Improve reward functions, agentic loops, custom tool integration
- New academic compute coming: Argonne's Solstice system with **100,000 NVIDIA Blackwell GPUs**